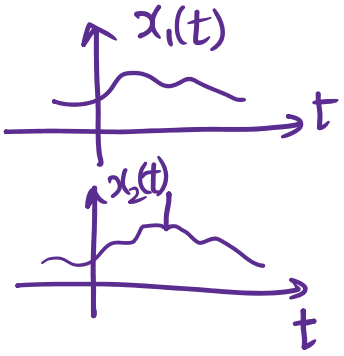
C.T.F.T.

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt \qquad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega) e^{j\omega t}\, d\omega$$

Properties:

$[x(t)] \longleftrightarrow [X(\omega)]$



1) Linear & Invertible

2) $x(t)$ real $\Rightarrow X(-\omega) = X^*(\omega)$

3) $x(t)$ is real & even $\Rightarrow X(\omega)$ is real & even

4) $x(t-t_0) \longleftrightarrow X(\omega) e^{-j\omega t_0}$ } Time Shifting

Pf: $\int_{-\infty}^{\infty} x(t-t_0) e^{-j\omega t}\, dt = \int_{-\infty}^{\infty} x(t') e^{-j\omega(t'+t_0)}\, dt'$, $t' = t - t_0$

$$= e^{-j\omega t_0} \int_{-\infty}^{\infty} x(t') e^{-j\omega t'}\, dt'$$

$$= X(\omega) e^{-j\omega t_0}$$

5) $x(t) e^{j\omega_0 t} \longleftrightarrow X(\omega - \omega_0)$

Pf: $\int_{-\infty}^{\infty} x(t) e^{j\omega_0 t} e^{-j\omega t}\, dt = \int_{-\infty}^{\infty} x(t) e^{-j(\omega-\omega_0)t}\, dt = X(\omega - \omega_0)$

6) $\frac{dx(t)}{dt} \longleftrightarrow j\omega X(\omega)$

Pf: $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega) e^{j\omega t}\, d\omega$

$$\Rightarrow \underbrace{\frac{dx(t)}{dt}}_{y(t)} = \frac{1}{2\pi}\int_{-\infty}^{\infty} \underbrace{X(\omega)\, j\omega}_{Y(\omega)} e^{j\omega t}\, d\omega$$

$\therefore\ y(t) \longleftrightarrow Y(\omega)$ i.e. $\frac{dx(t)}{dt} \longleftrightarrow j\omega X(\omega)$

7) $t x(t) \longleftrightarrow j \dfrac{dX(\omega)}{d\omega}$

<u>Pf:</u> $X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$

$\therefore \underbrace{\dfrac{dX(\omega)}{d\omega}}_{Y(\omega)} = \int_{-\infty}^{\infty} \underbrace{x(t)(-jt)}_{y(t)} e^{-j\omega t} dt$

$\therefore -jt x(t) \longleftrightarrow \dfrac{dX(\omega)}{d\omega} \Rightarrow t x(t) \longleftrightarrow j \dfrac{dX(\omega)}{d\omega}$

8) $x(at) \longleftrightarrow \dfrac{1}{|a|} X\left(\dfrac{\omega}{a}\right)$ } Scaling

Pf: $\int_{-\infty}^{\infty} x(at) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t') e^{-j\omega \frac{t'}{a}} \dfrac{dt'}{a}$, $at = t'$ $(a>0)$

$= \dfrac{1}{a} \int_{-\infty}^{\infty} x(t') e^{-j\left(\frac{\omega}{a}\right)t'} dt'$

$= \dfrac{1}{a} X\left(\dfrac{\omega}{a}\right)$, $a>0$

$a<0 \Rightarrow \dfrac{-1}{a} X\left(\dfrac{\omega}{a}\right)$

$\Big\}$ $\dfrac{1}{|a|} X\left(\dfrac{\omega}{a}\right)$

9) $\int_{-\infty}^{\infty} |x(t)|^2 dt = \dfrac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$ } Parseval's rel^n

Pf: $\int_{-\infty}^{\infty} x(t) x^*(t) dt = \int_{-\infty}^{\infty} x(t) \dfrac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) e^{-j\omega t} d\omega \, dt$

$= \dfrac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) d\omega \underbrace{\int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt}_{X(\omega)}$

$= \dfrac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$

10) $x(t) \longleftrightarrow X(\omega)$
$X(t) \longleftrightarrow 2\pi x(-\omega)$ } Duality Property

∞ $\quad$ ∞ , $-j\omega\tau$,

$$X(t) \longleftrightarrow 2\pi x(-\omega)$$

<u>Pf:</u> $X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau} d\tau$

$$\therefore x(t) = \int_{-\infty}^{\infty} x(\tau) e^{-jt\tau} d\tau = \int_{-\infty}^{\infty} x(\omega) e^{-jt\omega} d\omega \quad , \quad -\omega = \omega'$$

$$= -\int_{\infty}^{-\infty} x(-\omega') e^{j\omega' t} d\omega'$$

$$= \int_{-\infty}^{\infty} x(-\omega) e^{j\omega t} d\omega$$

$$\therefore \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi x(-\omega) e^{j\omega t} d\omega$$

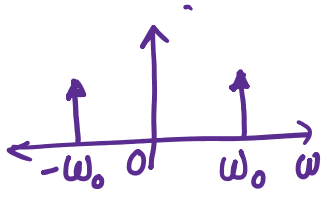
$$\therefore \quad x(t) \longleftrightarrow 2\pi x(-\omega)$$

<u>Ex:</u> 1) $\left.\begin{array}{l} \delta(t) \longleftrightarrow 1 \\ 1 \longleftrightarrow 2\pi\delta(-\omega) = 2\pi\delta(\omega) \end{array}\right\} \Rightarrow 1 \longleftrightarrow 2\pi\delta(\omega)$

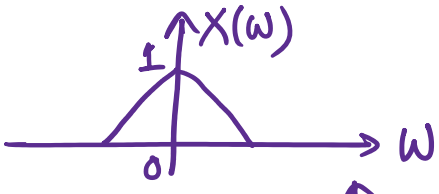
2) $1 \cdot e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - \omega_0)$

3) $\cos(\omega_0 t) \longleftrightarrow ?$

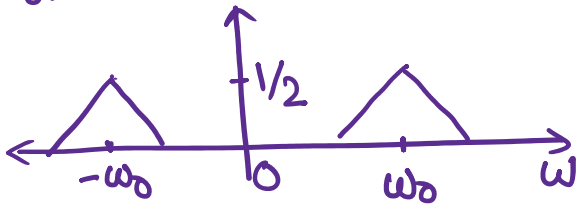
$$= \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \longleftrightarrow \pi\left[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)\right]$$



4) $\underbrace{x(t)}_{\text{message}} \cos(\overbrace{\omega_0 t}^{\text{carrier}}) \longleftrightarrow ?$



$$= x(t)\left[\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}\right] \longleftrightarrow$$



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11) $x(t) * h(t) \longleftrightarrow X(\omega) H(\omega)$

Pf: $y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = \int_{-\infty}^{\infty} x(\tau) \frac{1}{2\pi} \int_{-\infty}^{\infty} H(\omega) e^{-j\omega\tau} \cdot e^{j\omega t} d\omega$

Pf: $y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau = \int_{-\infty}^{\infty} x(\tau)\frac{1}{2\pi}\int_{-\infty}^{\infty} \ldots$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} H(\omega)e^{+j\omega t}\,d\omega \underbrace{\int_{-\infty}^{\infty} x(\tau)e^{-j\omega\tau}\,d\tau}_{X(\omega)}$$

$$y(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \overbrace{X(\omega)H(\omega)}^{Y(\omega)} e^{j\omega t}\,d\omega$$

$$\Rightarrow \quad y(t) \longleftrightarrow Y(\omega) = X(\omega)H(\omega)$$

$$\therefore \quad x(t) * h(t) \longleftrightarrow X(\omega)H(\omega)$$

12) $x(t)h(t) \longleftrightarrow \frac{1}{2\pi} X(\omega) * H(\omega) = Y(\omega)$

Pf: $\frac{1}{2\pi}\int_{-\infty}^{\infty} X(\rho)\underbrace{H(\omega-\rho)}\,d\rho = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\rho)\int_{-\infty}^{\infty} h(t)e^{j\rho t}e^{-j\omega t}\,dt\,d\rho$

$$= \underbrace{\frac{1}{2\pi}}\int_{-\infty}^{\infty} h(t)e^{-j\omega t}\,dt \underbrace{\int_{-\infty}^{\infty} X(\rho)e^{j\rho t}\,d\rho}$$

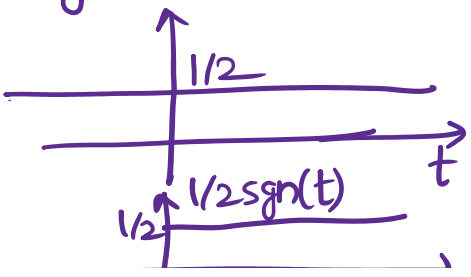
$$Y(\omega) = \int_{-\infty}^{\infty} \underbrace{x(t)h(t)}_{y(t)} e^{-j\omega t}\,dt$$

$\therefore y(t) \longleftrightarrow Y(\omega)$ i.e. $x(t)h(t) \longleftrightarrow \frac{1}{2\pi} X(\omega) * H(\omega)$

13) $\int_{-\infty}^{t} x(\tau)\,d\tau \longleftrightarrow ?$

$= x(t) * u(t) \longleftrightarrow X(\omega)U(\omega)$, $U(\omega)$?

$$U(\omega) = \int_{0}^{\infty} e^{-j\omega t}\,dt$$

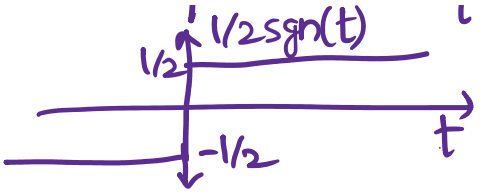


$$u(t) = \frac{\mathrm{sgn}(t)}{2} + \frac{1}{2}$$

$$\frac{\quad}{2} + \frac{\quad}{2}$$

$$\downarrow \qquad \updownarrow$$

$$? \qquad \pi\delta(\omega)$$



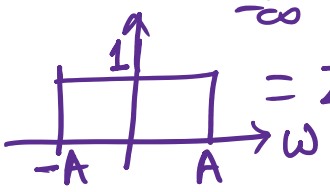
$$f(t) = \frac{1}{t} \longleftrightarrow F(\omega)?$$

odd f^n of t

$$F(\omega) = \int_{-\infty}^{\infty} \frac{1}{t} e^{-j\omega t}\, dt = \int_{-\infty}^{\infty} \frac{\cos(\omega t)}{t} - j\frac{\sin(\omega t)}{t}\, dt$$

$$= -j\int_{-\infty}^{\infty} \pi \frac{\sin(\omega t)}{\pi t}\, dt = -j\pi \underbrace{\int_{-\infty}^{\infty} \frac{\sin(\omega t)}{\pi t}\, dt}_{=1,\ \omega>0} = \underbrace{-j\pi}_{F(\omega)}$$

$$z(t) = \frac{\sin At}{\pi t} \longleftrightarrow$$



$$= Z(\omega) = \int_{-\infty}^{\infty} z(t) e^{-j\omega t}\, dt$$

$$\Rightarrow Z(0) = \int_{-\infty}^{\infty} z(t)\, dt = 1$$

for $\omega < 0 \Rightarrow j\pi = F(\omega)$, $F(\omega) = -j\pi$, $\omega > 0$

$$\therefore F(\omega) = -j\pi\, \mathrm{sgn}(\omega)$$

$$\therefore \frac{1}{t} \longleftrightarrow -j\pi\, \mathrm{sgn}(\omega)$$

$$\Rightarrow -j\pi\, \mathrm{sgn}(t) \longleftrightarrow 2\pi \frac{-1}{\omega} = \frac{-2\pi}{\omega}$$

$$\Rightarrow \mathrm{sgn}(t) \longleftrightarrow \frac{2}{j\omega}$$

$$\therefore \frac{\mathrm{sgn}(t)}{2} \longleftrightarrow \frac{1}{j\omega}$$

$$\therefore U(\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$$

$$\therefore \int_{-\infty}^{t} x(\tau)\, d\tau \longleftrightarrow \frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)$$

$$x_1(t) = 5, \quad x_2(t) = 10$$

$$\underbrace{x_1'(t) = 0}_{\updownarrow 0} \qquad \underbrace{x_2'(t) = 0}_{\updownarrow 0}$$

$$\underbrace{x_1(t) = 0}_{\updownarrow\, 0} \qquad \underbrace{x_2(t) = 0}_{\updownarrow\, 0}$$